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Computer Science Genre Report

Scrum Product Backlog

In the field of computer science, project teams (at a company, for example) adhere to software development methodologies. These methodologies prescribe certain processes that divide the project's workflow into steps. There are standard, well-defined methodologies in frequent use; one of these is Scrum, created by Ken Schwaber and Jeff Sutherland. Scrum is a methodology with an Agile process, so it is often used when requirements may change quickly. (Aristek Systems)

The Scrum process has many elements – roles, work cycles, and artifacts. Artifacts are deliverables that facilitate and document the project workflow. These can be texts, meetings, or project components. One textual artifact in Scrum is the Product Backlog. This is a prioritized list of requirements to be implemented. Each work cycle (called a Sprint), some of these features are chosen to be worked on. Within Scrum practices, the Product Backlog is not intended to be comprehensive when created; it is a changing document that adjusts to the needs of the project's development. For example, it may be updated as bugs are discovered, or as new ideas emerge. (Schwaber & Sutherland)

In Scrum, there are 3 main roles – Product Owner, Scrum Master, and Developers. The product owner is typically a boss or client. The Scrum Master is like a team leader. And the developers implement the software. The Product Owner is mainly responsible for creating and maintaining the Product Backlog, but they will receive continuous input on this from the Scrum Master and Developers. (Scrum Organization)

The Product Backlog is internal to the project team (defined in the Scrum sense as Product Owner, Scrum Master, and Developers), and the audience is the whole project team. The Product Owner will reference the Product Backlog they maintain to keep track of process – past, present, and future. The Scrum Master uses the Product Backlog to select, with input from Product Owner and Developers, what requirements will be implemented in a particular Sprint. The Developers use the Product Backlog to understand details about the requirements they are implementing; (Schwaber & Sutherland) for example, the developers may need to consider constraints on compatibility when implementing a certain requirement. If the Product Backlog is poorly maintained (e.g. unclearly specified requirements), the process will be untransparent and therefore inefficient.

The Product Owner uses the Product Backlog to promote efficient progress towards a high value product. (Schwaber & Sutherland) The other members of the team seek this from the Product Backlog. Trust and transparency are considered important in Scrum, so it should be noted that Product Owner, Scrum Master, and Developers are idealized to have a common goal for the project and work towards this together. As such, the Product Backlog is for everyone's use in efficient progress towards a high value product, though those of different roles will use it differently as I've described.

The creators of Scrum do not provide guidelines on formatting, or even medium. They state that the Product Backlog is an “emergent, ordered list”. (Schwaber & Sutherland) So, it changes, and it is in order of precedence. This may look like an electronic text document, the most intuitive option, or perhaps a handwritten list on a whiteboard.

The Product Backlog will likely use technical language for specificity, but ideally the artifact will be written with openness and the common goal in mind, so meaning will be clear to the audience (the Scrum team). Diagrams may be used to visualize designs or processes. For example, a sequence diagram may show how a user-interaction should play out. (Bell) Sources are not relevant to Product Backlogs.

As mentioned, listed requirements will be requirements that can be implemented in one Sprint; requirements may need to be refined (broken into smaller requirements) to meet this. Requirements will vary between projects. For example, a video game’s Product Backlog may have control requirements, level design, and graphical bug fixes as some requirements. A website’s Product Backlog may have style elements and form problem fixes as listed requirements.

Scrum emphasizes certain values – foremost Trust and then Courage, Focus, Commitment, Respect, and Openness. (Schwaber & Sutherland) Trust refers to trust that all members are working towards the common goal of product completion. This is an idealization; the goals of all Scrum team members cannot be expected to be perfectly aligned. For example, the Developers and Scrum Master may be seeking a raise or promotion, and their efforts toward that goal may conflict with efficient completion of the project; perhaps a developer takes upon more tasks for themselves to look good at their job, even though it slows the project’s workflow down. The Scrum values are important for the methodology to work, but it should also be

recognized when there are conflicting motives in the team. When this happens, it may be appropriate to make this conflict known amongst the team or even resolve it.

Works Cited

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